

TIC 401936263

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.201021 d
Epoch (t0) 1517.7707 BTJD
Duration 0.1700 d
Depth 5.787 % (57,875 ppm)
SNR 271.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 22 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 164.0 (gate ≥ 7.0)
Odd/even z-score 1.74 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 120.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 82.33 (gate ≥ 5.0) Compactness 0.96 (gate ≥ 0.50)
Dimming centroid offset 0.06 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 401936305
2.14 px away, dTmag +1.65 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

